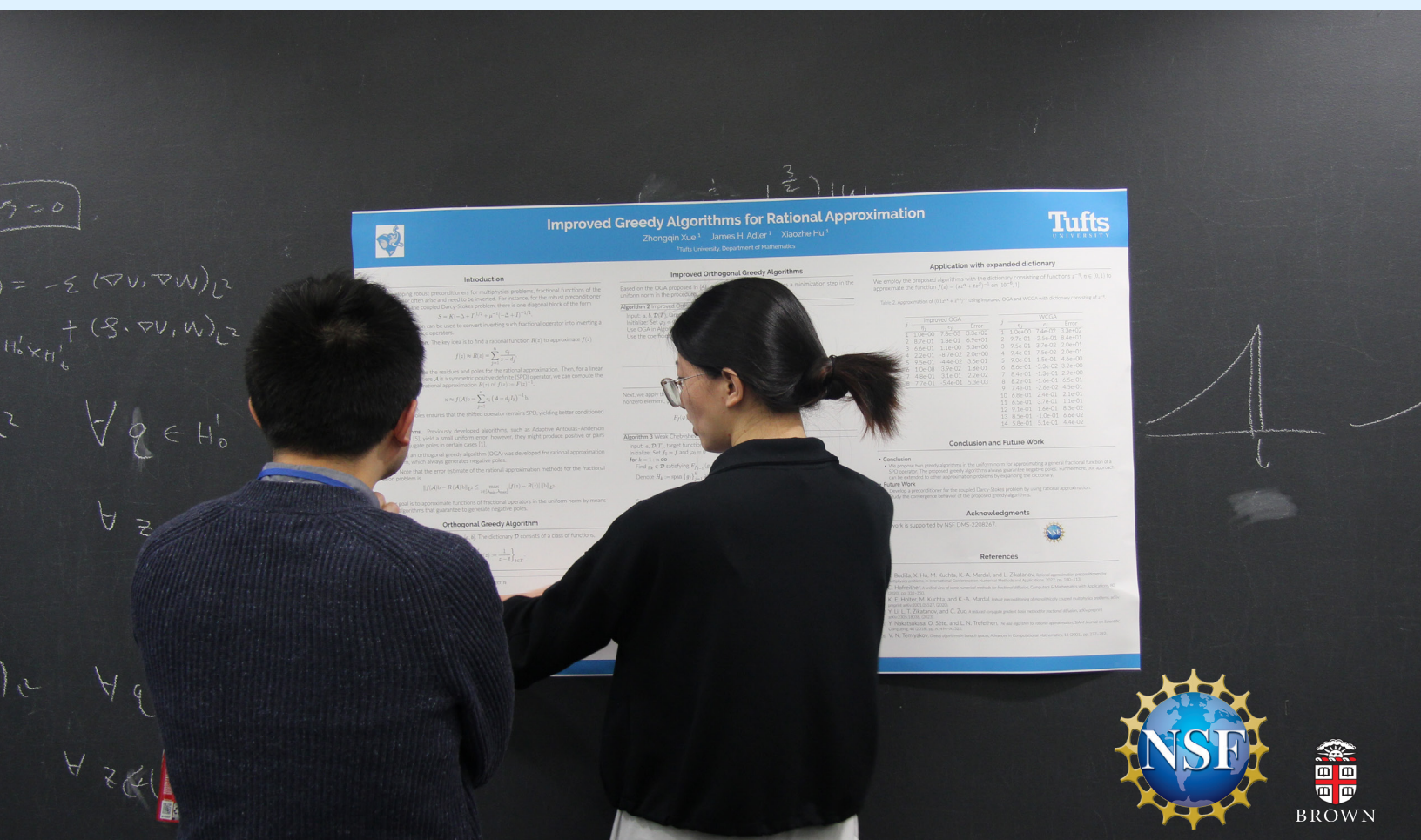




Spring 2024 Newsletter



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Greetings!

This Spring, ICERM has been home to the Semester Program [Numerical PDEs: Analysis, Algorithms, and Data Challenges](#). The program is co-organized by [Marta D’elia](#) (Pasteur Labs and Stanford University), [Johnny Guzman](#) (Brown University), [Brittany Hamfeldt](#) (New Jersey Institute of Technology), [Michael Neilan](#) (University of Pittsburgh), [Maxim Olshanskiy](#) (University of Houston), [Sara Pollock](#) (University of Florida), [Abner Salgado](#) (University of Tennessee), and [Valeria Simoncini](#) (Università di Bologna).

So far this semester, the workshops [Numerical Analysis of Multiphysics Problems](#) and [PDEs and Geometry: Numerical Aspects](#) have brought nearly 200 researchers to ICERM to explore the latest trends in computational PDEs, with a focus on developing new numerical methods and analyzing their efficiency. During Numerical Analysis of Multiphysics Problems, we saw attendees develop exciting research accelerating Newton’s method near singular points and connecting preconditioning with acceleration. Other hot topics included nonlinearity and fluid-structure interactions. At PDEs and Geometry: Numerical Aspects, participants focused on optimal transport, data-driven approaches to understanding geometric evolutions, and fluid dynamics on surfaces. In April, ICERM will host the final spring semester workshop [Nonlocality: Challenges in Modeling and Simulation](#). This workshop will provide a forum for researchers to discuss the modeling, analysis, approximation, and recent applications of nonlocal equations.

ICERM also hosted the Simons Collaboration Workshop [K3 Surfaces](#) and the Hot Topics Workshop [The Industrialization of SciML](#). At The Industrialization of SciML, experts discussed recent advances in applied scientific machine learning and how to overcome obstacles to their implementation across the scientific industries. During K3 Surfaces, a small team of researchers worked to integrate information about families of K3 surfaces and their elliptic fibrations into the L-functions and Modular Forms Database (LMFDB). The [LMFDB](#), an open-access resource supported in part by the [Simons Collaboration on Arithmetic Geometry, Number Theory, and Computation](#), is a database of mathematical objects arising in number theory and arithmetic geometry. In 2022, elliptic curve datasets from the LMFDB helped mathematicians at the University of Connecticut uncover the murmurations phenomenon. You can read about their discovery in this [recent article](#) in *Quanta Magazine*. We look forward supporting future research with new additions to the database!

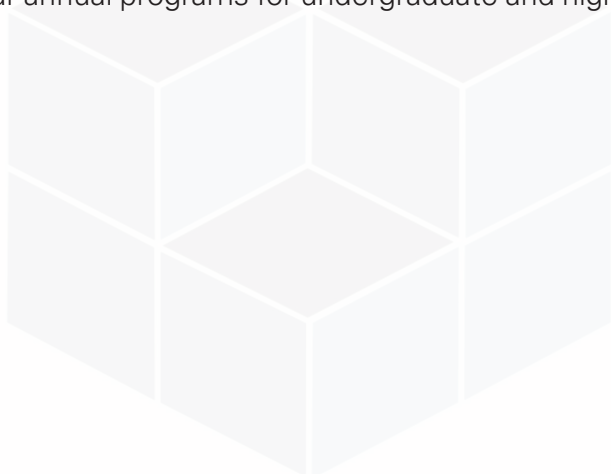
With summer approaching, we are in preparation mode for our busiest season of the year. Workshop season at ICERM will span topics from systems of interacting particles, to applications of random matrices, to simulating extreme spacetimes. We will also host several programs for early-career and historically excluded researchers to foster connections and gain critical research experience. Finally, we look forward to welcoming students for [Summer@ICERM](#) and [GirlsGetMath@ICERM](#), our annual programs for undergraduate and high school student-mathematicians.

We hope to see you at a program this summer!

Sincerely,



Brendan Hassett



UPCOMING PROGRAMS

Semester Programs

Fall 2024: [Theory, Methods, and Applications of Quantitative Phylogenomics](#)

Spring 2025: [Geometry of Materials, Packings, and Rigid Frameworks](#)

Fall 2025: [Categorification and Computation in Algebraic Combinatorics](#)

Spring 2026: [Stochastic and Randomized Algorithms in Scientific Computing: Foundations and Applications](#)

Topical Workshops

Jun 3-7, 2024: [Recent Progress on Optimal Point Distributions and Related Fields](#)

Jun 24-28, 2024: [Queer in Computational and Applied Mathematics \(QCAM\)](#)

Jul 8-12, 2024: [Solving the Boltzmann Equation for Neutrino Transport in Relativistic Astrophysics](#)

Jul 15-19, 2024: [Braids Reunion Workshop](#)

Aug 5-9, 2024: [Simulating Extreme Spacetimes with SpEC and SpECTRE](#)

Aug 19-23, 2024: [Spectral Analysis of Schrödinger Operators](#)

Dec 9-13, 2024: [Harmonic Analysis and Convexity](#)

Hot Topics Workshops

May 20-22, 2024: [Random Matrices and Applications](#)

Sep 13-15, 2024: [Robust Optimization and Simulation of Complex Stochastic Systems](#)

UPCOMING OPPORTUNITIES

GirlsGetMath@ICERM 2024

Applications are open for GirlsGetMath, ICERM’s annual summer program for high school students! Rising 10th- and 11th-graders near or in the greater Rhode Island area are eligible to apply, regardless of gender. GirlsGetMath is a five-day non-residential program where students are introduced to topics such as cryptography, image processing, and data science. This year’s program will be held from **August 12-16, 2024**.

NSF Artificial Intelligence, Formal Methods, and Mathematical Reasoning (AIMing) Award

The AIMing program seeks to support research at the interface of innovative computational and artificial intelligence technologies and new strategies/technologies in mathematical reasoning to automate knowledge discovery. The next deadline for proposals is **June 3, 2024**.

NSF Partnerships for Research Innovation in the Mathematical Sciences (PRIMES) Award

The PRIMES program aims to support partnerships between [NSF Division of Mathematical Sciences-supported institutes](#) and faculty at [minority-serving institutions](#). The program seeks to boost the participation of members of underrepresented groups in the mathematical sciences through their increased involvement in research programs at the institutes. The next deadline for proposals is **August 21, 2024**.

Propose a Satellite Event for the 2026 International Congress of Mathematicians (ICM)

On behalf of the International Mathematical Union, the ICM invites proposals for Satellite Conferences and other possible events to be associated with and complementary to the 2026 ICM. Satellite events may be hosted at universities, research institutes, and scientific organizations across the United States, Canada, and Mexico. Events should occur between May and September 2026 and should involve an international perspective on some aspect of the mathematical sciences with conference themes relating to the scientific areas covered by the ICM. Visit the [ICM website](#) for details.

PROPOSE A PROGRAM

ICERM is accepting proposals for the following program types:

Semester Programs

ICERM hosts a Spring and Fall Semester Program every year. Each has organizers and long-term participants who are expected to be in-residence for the majority of the semester. Semester programs typically incorporate three week-long associated workshops. For a typical program, ICERM provides travel and lodging support for 5-10 organizers (at least a month in residence, with some for the full semester); 20-30 long-term participants (4+ weeks); 1 Institute and 5 Semester postdoctoral fellows (salaried); 60 short-term participants (1-4 weeks); 10-15 graduate students (6+ weeks); and workshop attendees. ICERM helps essential long-term participants negotiate sabbatical leaves or teaching releases to foster their participation.

Summer@ICERM

Summer@ICERM is a Research Experience for Undergraduates led by faculty organizers who develop a variety of topical research projects that are interdisciplinary in nature. Throughout the eight-week program, students work on their assigned projects in groups of two-to-four, mentored by faculty advisors and teaching assistants. Summer@ICERM is not your typical REU, in that ICERM provides comprehensive start-to-finish logistical support to the faculty organizers, including assistance recruiting TAs. In addition, faculty and TAs are paid summer salary with lodging support, and the average faculty/TA-to-student ratio is 1:3.

Topical Workshops

Topical Workshops run over 5 weekdays and focus on a topical theme that aligns with the Institute’s mission of supporting and broadening the relationship between mathematics and computation. ICERM hosts 5-7 topical workshops each year. They are typically scheduled in December, January, and May through August (around the dates of the semester programs). ICERM covers travel and lodging expenses of the organizers and 20-25 invited speakers/lead participants; some participants (applicants) to the workshop may also be supported.

Graduate Training Workshops

Graduate Training Workshops are 1-3 week events on a timely and exciting mathematical or computational theme, with a focus on training graduate students, recent PhDs, and researchers new to the field. They are typically scheduled in December, January, and May through August (around the dates of the semester programs). ICERM covers travel and lodging expenses of the organizers, invited speakers, TAs, and early-career participants.

Collaborate@ICERM

Collaborate@ICERM offers teams of 3-6 researchers the opportunity to spend five days at the Institute during the summer (May-August) or during the month of January. The team research project should have a computational or experimental component. ICERM provides access to a variety of software packages and high performance computing through Brown’s [Center for Computation and Visualization](#). Proposals involving research projects initiated at past ICERM programs are encouraged. This program provides limited funding for travel to the Institute and local accommodations for six nights.

Hot Topics Workshops

Hot Topics Workshops are designed to allow ICERM to move quickly in order to start the public exploration of breakthroughs and emerging mathematical areas. They run 2-3 days and are organized on a few months’ notice. They can originate through suggestions from ICERM boards or the community.

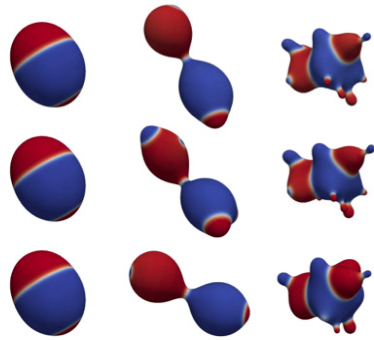
Deadlines and Review Process

Hot Topics Workshop suggestions are reviewed on a rolling basis. The next target deadline for all other proposal types is **May 1, 2024**. Submissions and inquiries should be sent to [ICERM’s Director](#). ICERM’s Directorate and the Scientific Advisory Board (SAB) review all proposals. Proposers will receive feedback within a month of the SAB meeting.

For more information, visit icerm.brown.edu/proposals.

Stepping Inside the Cahn-Hilliard Equation

[PDEs and Geometry: Numerical Aspects](#) co-organizer [Dr. Axel Voigt](#) (*Technische Universität Dresden*) combines mathematics, computation, and design to create immersive, interactive research spaces. In 2016, Voigt presented the exhibit in the video below in collaboration with Dr. Florian Stenger, THÉORIZ Studio, and Technische Universität Dresden. The piece illustrates the dynamics of the Cahn-Hilliard equation, allowing users to explore simulated bicontinuous structures by combining fluid simulations with rendering and tracking technology.



Voigt: The evolution is driven by the surface tension of the interface of two fluids, which results in a continuous reduction of the interface area and leads to self-similar structures. The process is mathematically described by a Cahn-Hilliard equation, numerically solved by an adaptive Finite Element Method on the high-performance computer JURECA at the [Jülich Supercomputing Centre](#). In each time step the structure, visualized by the fluid-fluid interface, or slices of it, is rendered. According to the position of person the corresponding time frame is locally interpolated and projected. By moving through the room one therefore can interact with the structure and evolve it in time in a playful and intuitive manner.

Video courtesy of Axel Voigt.

Cahn-Hilliard phase field models were the topic of several lectures during the Semester Workshops [Numerical Analysis of Multiphysics Problems](#) and [PDEs and Geometry: Numerical Aspects](#). Visit ICERM’s [video archive](#) to access talks from the Semester Program!

Monica Stephens: Building Bridges Between Spelman and ICERM

Stephens is a long-term participant in ICERM's Spring 2024 Semester Program and a U.S. National Science Foundation Partnerships for Research Innovation in the Mathematical Sciences (PRIMES) awardee. The PRIMES program seeks to support research partnerships between NSF Division of Mathematical Sciences-supported institutes and faculty at [minority serving institutions](#), including historically Black colleges and universities, Hispanic-serving institutions, Tribal colleges and universities, and Asian-American and Pacific Islander-serving institutions. Learn more on the [program website](#).



For Dr. Monica Stephens (*Spelman College*), ICERM's Spring 2024 Semester Program [Numerical PDEs: Analysis, Algorithms, and Data Challenges](#) is the next step in a career-long commitment to creating opportunities for historically excluded mathematicians. A recipient of one of the first National Science Foundation Partnerships for Research Innovation in the Mathematical Sciences (PRIMES) awards, Stephens is spending the semester at ICERM conducting research and developing course curriculum to bring back to Spelman College, a historically Black women's college in Atlanta, Georgia.

Stephens, now an Associate Professor of Mathematics, was first encouraged to study math as a Spelman undergraduate. After entering the College as an engineering major, Stephens took a course with renowned mathematician and educator Dr. Etta Zuber Falconer, one of the first Black women to earn a mathematics Ph.D. in the United States. "I discovered through my class with her that I really enjoyed mathematics, and to see her and the other women in the department who were doing it, they're the ones who really planted the seed for me that I should think about doing math and should think about getting a Ph.D.," she shared.

Going on to earn her master's and doctoral degrees from Brown University, Stephens recalls a difficult social transition into graduate school. Coming from an all-women's institution, she was drawn to Brown by the significant number of women in the Applied Mathematics department at the time. Still, making the transition from an HBCU proved to be a challenge. "I remember as a student having to find other ways to find community because I didn't always find it in my department. I was fortunate to have quite a few women in the department so we had a group, but there were no other Black folks in the department, no Black instructors in the department, so I ended up finding community in other ways." Nevertheless, Stephens "absolutely felt supported as a graduate student in [Brown's] Applied Math department."

Her return to Providence has been a chance to focus on research after years of shouldering a heavy teaching load, serving as chair of the math department, and coordinating the [WiSTEM Summer Bridge Accelerator program](#) at Spelman, a primarily undergraduate institution. "At a teaching institution, a lot of times faculty don't have a huge amount of time during the academic year to do research work. They may be able to engage and do a little bit of work, but we're teaching six courses a year," Stephens explained. "I feel like I'm investing this time in myself." With the time afforded by her PRIMES award, Stephens has been working with ICERM Deputy Director Emerita Sigal Gottlieb to develop computational techniques to solve equations modeling spinodal decomposition, the separation of binary mixtures into two phases. "What we are trying to do is develop numerical methods that accelerate the convergence of the method or come up with techniques to accelerate the convergence of the method, and then make sure that we're analyzing in detail whether or not these methods are stable."

While in-residence at ICERM, Stephens continues to think of ways to share the benefits of her award with students and colleagues at Spelman. "We have a numerical analysis course, and I would like to do some curriculum development and infuse some numerical PDEs into it so students have that exposure," she shared. She emphasized that at mathematical research destinations like ICERM, "working with junior faculty at HBCUs, providing summer opportunities for faculty, and having opportunities for undergraduates to engage [are important]." Leading by example, Stephens included opportunities for two Spelman undergraduate students and Professor Erdi Kara to visit ICERM in her PRIMES proposal. In March, the group attended [The Industrialization of SciML](#), an artificial intelligence-focused Hot Topics Workshop.

Now approaching the end of her residency, Stephens has experienced firsthand how attending a long-term program can shape participants' research for years to come. "As I go to conferences, when I look at the landscape of mathematics it is still very much lacking diversity," she remarked. She hopes PRIMES and future programs will help to change that. In her own classroom, Stephens frequently tells undergraduate students that by pursuing graduate school, they can help make a path for others to do advanced study in mathematics. "For those of us who've made it," she believes, "we have a responsibility to reach back and bring along others who are coming along the path." ■

"Numbers don't lie": Translating Community Storytelling to Data Storytelling by [Carrie Diaz-Eaton](#), [Joseph Hibdon](#), and [Kiara Sanchez](#)

During the 2023 [From Impact Factor to Influence Factor: Data Science and Policy for Social Justice](#) workshop at ICERM, Dr. Selena Fisk, Dr. Mickey McGlasson, and others spoke about the importance of data storytelling. In her book *I'm Not a Numbers Person: How to Make Good Decisions in a Data-Rich World*, Dr. Fisk explains that "data storytelling involves thinking about what the numbers tell you about real people and real lives, and considering what your actions should be as a result of knowing that information." Data storytelling is of critical importance because, if we aim to use data to effect change, we must be able to communicate our insights with a wide audience, including community members, advocates, and politicians, who may not have a quantitative background.

During the workshop, we worked with local Providence residents to ask what issues matter to them. For optimal impact, our team of researchers partnered with community organizers from the [Woonasquatucket River Watershed Council's](#) (WRWC) *Nuevas Voces* leadership program. They were able to provide us with real and timely issues that their communities were facing, and for which they needed access to data. Of dire concern was access to proper healthcare. In particular, residents were unaware of what services were provided at which facilities, and where they would be welcomed.

While gathering data, we learned that almost 40% of Providence is Spanish-speaking. Through community member voices and the work of Dr. Pablo Rodriguez, our team learned that there are not enough Spanish-speaking medical providers to serve the large Spanish-speaking population in Providence. Our team was able to leverage our expertise as researchers and data scientists to quantify this problem. We found that the ratio of Spanish-speaking residents to Spanish-speaking doctors is approximately 191 to 1, while the national average is approximately 22 to 1.¹

The lack of Spanish-speaking medical providers is not the only hindrance to the community in receiving proper medical care. In our quest to understand the landscape of medical services for the Providence community, we discovered that a large population of residents have been barred from receiving medical attention. That is, adult immigrants in Rhode Island have virtually no options for health insurance. (*continued on page 13*)

¹ Hibdon, J. Ortega, O. (2023, August). *Language barrier a bar to good health. What can RI do*. The Providence Journal.

MEET ICERM'S SPRING 2024 POSTDOCTORAL FELLOWS*

John Carter Semester Postdoc

John Carter received his Ph.D. in 2023 from Missouri University of Science and Technology under the supervision of Daozhi Han. Following the completion of his program, he accepted a postdoctoral fellowship with Rensselaer Polytechnic Institute in 2023 working with Mark Shephard and Jefferey Banks. In January 2024, he joined ICERM as a Semester Postdoctoral Fellow in the Numerical PDEs: Analysis, Algorithms, and Data Challenges Semester Program. Carter's research interests include numerical analysis, finite element methods, data assimilation, and ensemble methods related to uncertainty quantification. In particular, he has studied applications of an ensemble mean that involve multiple realizations of parameters to develop high-order numerical schemes.



ICERM: What most excites you about what you're working on right now?

Carter: The application. The project I'm working on now is meant to help with modeling space weather. They're using multifluid plasma equations to try to do this. The physics is beyond me—I'm more of a numerical person—but it's interesting stuff, and the idea that they can use these equations to actually model these space phenomena is interesting.

ICERM: How would you describe your work to someone with little or no math background?

Carter: What I just described is kind of the overview of the project in general. But I'm mathematical, a more numerical person, so what I have been doing is developing the methods that they use to solve these equations to make these models happen. It's a combination of methods—one method meant to help solve for the magnetic and electric fields in the problem and the other meant to help solve for the state variables, like the density of the momentums and the energies involved in the equation. Then, these two methods are coupled together by another method, all of this written in C++. So learning C++ is probably more intimidating than actually developing the methods themselves I think.

ICERM: What first drew you to becoming a mathematician and what keeps you interested?

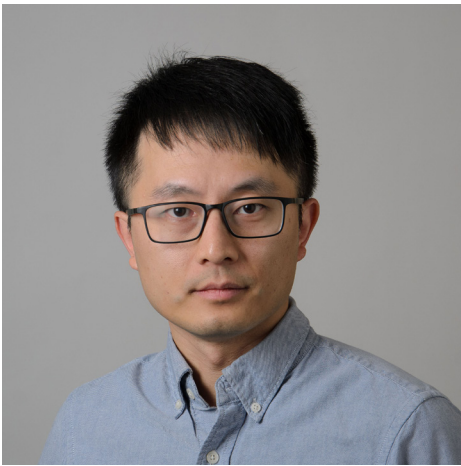
Carter: When I first got into math, I was actually in pure math. My master's degree is actually in pure math. So for me, it's sort of this kind of obsessive quality about the problems that come up with pure math. Back then, there wasn't any clear application. It's just fun for my mind to work with these problems. I did combinatorial geometry back then, which is a lot of organizing, finding patterns, and making connections. And it's just something I guess my mind likes.

ICERM: What are you looking forward to in the semester at ICERM?

Carter: The workshops really that have already occurred have been pretty fun. I've enjoyed them, so as long as more keep coming I'm happy about it. I have some colleagues I know who are coming, so I'm curious to meet with them and see what updates they have and what they've been working on, and maybe collaborate. Collaboration, I guess, is the biggest thing that I'm looking forward to. ■

Sijing Liu Institute Postdoc

Sijing Liu received his Ph.D. at Louisiana State University in 2020 under the supervision of Prof. Susanne C. Brenner. During 2020-2023, he was an Assistant Research Professor at the University of Connecticut working with Prof. Dmitriy Leykekhman. He joined ICERM as an Institute Postdoc in August 2023. His research lies in the fields of numerical analysis and scientific computing. Specifically, he is interested in finite element methods, multigrid methods, discontinuous Galerkin methods, PDE-constrained optimizations, and optimal control problems. His current work focuses on designing and analyzing finite element methods, discontinuous Galerkin methods, and multigrid methods for elliptic optimal control problems. He is also working on stabilized finite element methods for convection-dominated problems and interior estimates of finite element methods for singularly perturbed parabolic equations.



ICERM: What questions have you been working on since arriving at ICERM?

Liu: I have been working on various projects since arriving at ICERM. These include designing high order time-stepping methods for fluid-structure interaction problems with my mentor Professor Johnny Guzman, designing and analyzing dual-wind discontinuous Galerkin methods for convection-dominated problems with my collaborators, and developing HDG methods for optimal control problems with BDDC preconditioner. I have also initiated multiple projects with new collaborators and have revised and published [one paper](#) while at ICERM.

ICERM: How did working alongside the Fall 2023 Math + Neuroscience program influence your research?

Liu: It was a great experience to interact with people from different research areas. I spoke with several participants in the Math + Neuroscience program who have questions about numerical PDEs. This provided me with insights into what other people wish to solve using the tools and methods we have developed in our research fields. I believe this is fantastic, as it will generate new research in both fields. Identifying new problems with real-world impact is always the most important aspect of our research endeavors.

ICERM: Are there any specific applications or larger scientific or social problems that motivate your research?

Liu: Numerical PDEs, in general, play an important role in today's society. From weather forecasting and reservoir simulation to aircraft design, aerodynamics, and astronautics, the applications are endless in our world. A fundamental understanding of the numerical algorithms behind all these applications is pivotal for designing new algorithms and analyzing existing ones. My research focuses precisely on the theoretical side of the story. The development of new algorithms, combined with hardware breakthroughs, leads to a more efficient world. I hope my research can make an impact on the real world, and this has always been my motivation.

ICERM: What are you looking forward to in your second semester at ICERM?

Liu: I am thrilled to have the opportunity to engage with many top experts in our research field during this semester. Being able to discuss problems, ask questions, and share ideas in this setting is a truly unprecedented opportunity for young researchers like me. Not only can I gain insights into new research areas, but I can also build connections in the long run. The workshops are also exciting since these workshops attract even more experts from all over the world and will make a huge difference in my research and career in general. ■

Marissa Masden
Institute Postdoc

Marissa Masden received her Ph.D. in 2023 from the Mathematics department at the University of Oregon under the supervision of Dev Sinha. She joined ICERM as an Institute Postdoc in August 2023, and was a participant in the Math + Neuroscience Fall Semester Program. Her dissertation work focused on uncovering statistics about topological invariants of neural network functions using algebraic constructions from oriented matroid theory and geometric ideas from piecewise linear Morse theory, with the goal of characterizing the topological expressiveness of these functions. More generally, she is also interested in applications of computational topology, geometry, and combinatorics in the natural and mathematical sciences.



ICERM: What questions have you been working on since arriving at ICERM?

Masden: I have been working on several projects, many of them collaborative. In the direction of topological properties of neural network functions, I have been working on generalizing tools from discrete Morse theory to the polyhedral complex setting of ReLU neural networks, as well as developing further combinatorial characterizations of their decomposition of input space. I have also been applying computational tools I developed during my dissertation to the topological analysis of dynamical systems. Finally, I've also been working on questions about the classification capacity of a certain class of function inspired by last semester's program.

ICERM: How did working with alongside the Math + Neuroscience program last semester influence your research?

Masden: I'm really excited by the myriad of geometric and topological tools currently being used towards understanding neuroscientific data. Because the area of machine learning I focus on is primarily artificial neural networks, many of the structures I study are analogous, so I've been happy to broaden my scope and perspective. I've also started collaborating on projects with some of the other program participants.

ICERM: Are there any specific applications or larger scientific or social problems that motivate your research?

Masden: I am motivated by wanting to know why machine learning models work, and finding guarantees for their behavior. In my opinion, this is essential to be able to safely apply these models in settings where we need strong guarantees about the outputs, for example, for any safety-related application.

ICERM: What are you looking forward to in your second semester at ICERM?

Masden: It's great to have the ability to spend more of my time digging into some of the questions that I mentioned. The resources here and the expertise of those around me are unparalleled. ■

Henry von Wahl
Semester Postdoc

Henry von Wahl received his Ph.D. in 2021 from the University of Magdeburg, where he worked under the supervision of Thomas Richter. During 2022-2023, he was a Postdoc at the University of Vienna and since then has become a Lecturer at the University of Jena. He joined ICERM as a Semester Postdoc in January 2024, and is a participant in the Numerical PDEs: Analysis, Algorithms, and Data Challenges Semester Program. His research interests are the development and numerical analysis of computational methods for (multi-)physics problems modeled by PDEs. In particular, his focus is on unfitted finite element methods and discontinuous Galerkin methods.



ICERM: What most excites you about what you're working on right now?

Von Wahl: The problem I'm working on right now is something that some other people have already looked at but in a way that I would say isn't particularly satisfying. We're working on a Crank-Nicolson-type time-stepping scheme for a time-dependent PDE on a moving domain. The reason why the stuff before wasn't so satisfying was that it's a difficult problem, so it's exciting to now have the time to look at these difficult problems. We've got somewhere in some places already and haven't got somewhere in other places, but it's exciting to see where this journey will take us and if we can get these problems fixed.

ICERM: How would you describe your work to someone with little or no math background?

Von Wahl: Numerics generally is about solving problems on a computer, so what I do essentially has to do with trying to simulate physics on a computer, so getting a computer to understand how to solve physics-based problems in a way that you can show that the results will be good results that somehow mirror what is going to happen in reality.

ICERM: What first drew you to becoming a mathematician and what keeps you interested?

Von Wahl: I guess I was playing with numbers and this kind of thing from a very young age. Then, when I moved from Germany to the UK, working

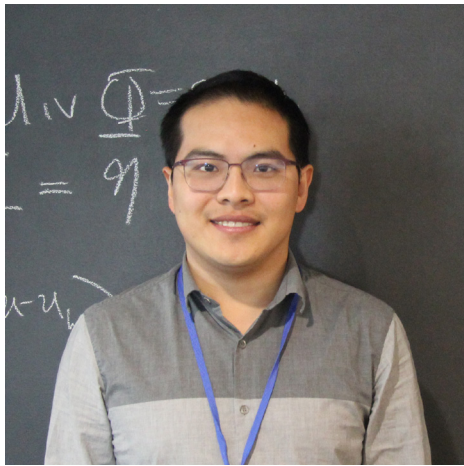
with numbers and with math wasn't as difficult because there isn't such a big language barrier with math. So I was always interested in it and I guess quite good at it. And it's just always fun. I mean it's always challenging, and it's particularly satisfying once you then manage to solve a difficult problem and get somewhere further. It is difficult in some sense, but you always get a larger reward once you solve something.

ICERM: What are you looking forward to in the year at ICERM?

Von Wahl: There are lots of great people who are here either the entire time or just visiting for one of the workshops, so I'm really looking forward to spending such long and intensive time with lots of great people and hopefully working together on some great new things. ■

Yukun Yue
Spring Postdoc

Yukun Yue received his Ph.D. in Mathematics from Carnegie Mellon University in 2023 under the guidance of Prof. Franziska Weber and Prof. Noel Walkington. Following his graduation, he joined the University of Wisconsin-Madison as a Van Vleck Visiting Assistant Professor in Fall 2023, collaborating with Prof. Qin Li. He is set to continue in this role through 2026, with a term as a Semester Postdoc at ICERM in Spring 2024. His primary research areas are computational mathematics and numerical analysis, with a focus on PDE-related problems emerging from physics. Currently, he is involved in projects related to liquid crystals simulation, computational fluid mechanics, and plasma control. He is also open to exploring other physics-driven problems.



ICERM: What most excites you about what you’re working on right now?

Yue: The most exciting part is that I can see that my work actually has some real applications. For example, my current work focuses on some optimization problems to find control for plasma. Plasma fusion and fusion energy are very big areas and people are interested in them, and I can see that my work and the construction of the mathematical model would actually help engineers and physicists in their research. This is just one example, actually. In addition to this, I would say that in many different areas as a mathematician, I think the important work for us is to construct the mathematical model and detect some rules from these models and they would help other sciences to improve, which makes me feel excited.

ICERM: How would you describe your work to someone with little or no math background?

Yue: So I think the first thing is that my job, my work is to let people not need to worry about math. There are many applications in math but usually, you need math as a tool to find some rule, and what actually benefits the world is the final result they have. So my job is basically to use computational measures and to use some theoretical math techniques to help people find those final results. They just need to use that final result, they do not need to worry about the process. And my job is to show the simulation to them to make it more understandable to other people.

ICERM: What first drew you to becoming a mathematician and what keeps you interested?

Yue: When I was still young, my primary interest was in physics, not in math. I wanted to be a physicist, however, I found that I do not have any gift for doing experiments. And then I discovered that what actually interests me in physics is setting up mathematical models and using mathematical deductions to discover physical rules and understand the world better. That’s what first interested me, and so I chose math as my major. All my work is still closely related to physics because it’s something I’m really interested in. I can have a lot of different collaborations with other people in that area as long as I’m a computational mathematician. I can collaborate with other people and, using our strengths together, we can discover something interesting.

ICERM: What are you looking forward to in the semester at ICERM?

Yue: The best thing about the program is it brings people from different areas and different universities together. I’ve already gotten to know a lot of people here and will get to know more in the future. I can learn the different areas people are interested in and get to know the tendencies of what they are working on, and then discuss potential areas where we can work together. That’s something you cannot have if you just stay at your home university. ■

“Numbers don’t lie”: Translating Community Storytelling to Data Storytelling
(continued from page 7)

We also found that even though there is technically state-sponsored insurance available for immigrant children in RI, undocumented parents are hesitant to risk revealing their own undocumented status by enrolling their children.² These findings demonstrate how quantitative researchers can gain critical knowledge outside of their disciplines through data storytelling projects.

Including the WRWC Nuevas Voces leaders in our work allowed us to reflect critically at each step of the data analysis process. They provided invaluable context from their own lived experiences and those of their families and friends, which continue to inform our work. To address the issue of Spanish-speaking residents not knowing where to seek medical services, our team built a healthcare map as part of an online data visualization tool project VECINA (Visualizing Environmental and Community Information for Neighborhood Advocacy), Spanish for “neighbor.” VECINA was developed at the [Institute for Mathematical and Statistical Innovation](#) (IMSI) in Chicago. The new Healthcare VECINA tool was included during our time at ICERM in the summer of 2023 and enables users to see healthcare facilities and resources on a map with detailed information about the medical services provided, in both English and Spanish (Figure 1). To reduce barriers due to illiteracy, the tool has an audio option where the user can also listen to the information about each location.

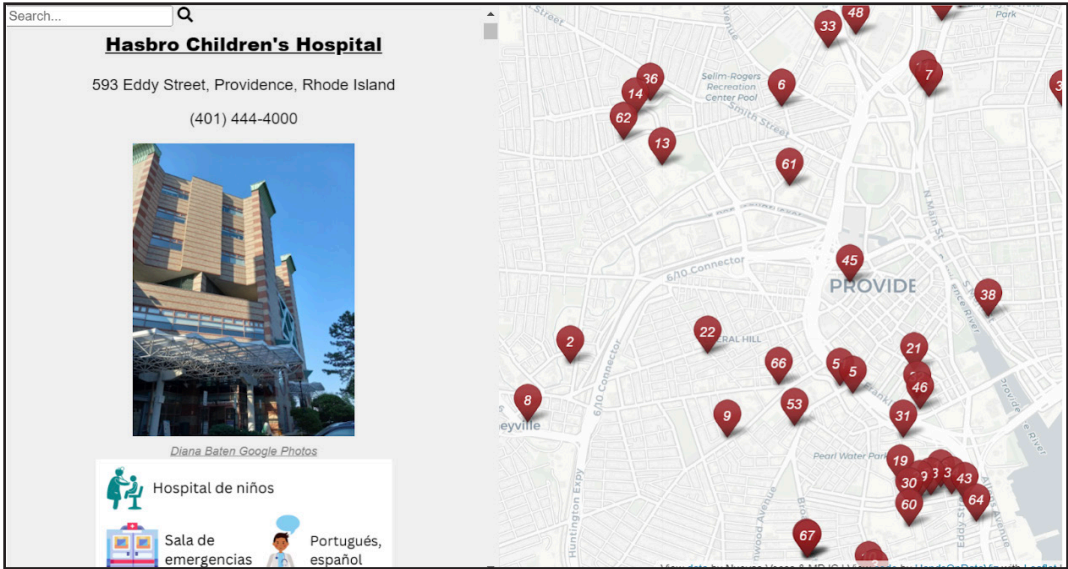


Figure 1: An image of the VECINA tool, showing medical facilities in Providence, RI. The panel on the left displays information about each facility when its marker is clicked.

VECINA will do more than point users to the appropriate medical facility—it will also provide a landscape of the health services in Providence, allowing community advocates and policymakers to fully understand the current reality and where they should concentrate their efforts.

The next steps are to work with WRWC to incorporate VECINA and other tools into their community education programming. We are also working to develop a website that will house the VECINA data tools and information in both Spanish and English languages. The team has met in person about twice per year for the last two years, switching between holding working groups in Providence at ICERM and in Chicago at IMSI. [Our next meeting is in May at IMSI](#). These intensive work sessions help the team design and refine the data infrastructure and data storytelling solutions, directed by community members’ needs. ■

2 Sanchez, K. (2023, August). *Rhode Island’s immigrants in dire need of health insurance*. The Providence Journal.

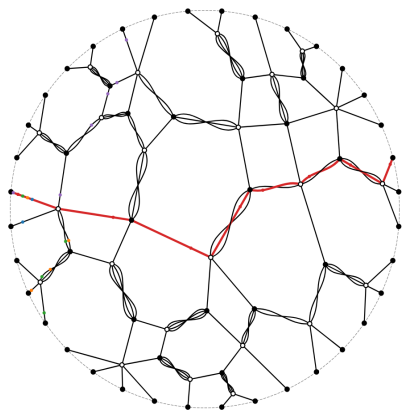


Photo courtesy of Joshua Swanson. Screenshot of the group's user interface for visualizing plabic graphs.

Web Bases, Promotion, and Plabic Graphs

From January 22-26, [Christian Gaetz](#) (Cornell University), [Oliver Pechenik](#) (University of Waterloo), [Stephan Pfannerer-Mittas](#) (Technische Universität Wien), [Jessica Striker](#) (North Dakota State University), and [Joshua Swanson](#) (University of Southern California) visited ICERM for the Collaborate@ICERM program Web Bases, Promotion, and Plabic Graphs. The project traces back to a previous Collaborate program in May 2023, where the group began a [series of papers](#) extending pre-existing work on three-row tableaux to the four-row case and describing its relation to the algebraic side of representation theory and special linear groups.

ICERM: Can you give a brief explanation of your Collaborate topic?

Pechenik: The title of our work is “Web bases, Promotion, and Plabic Graphs,” so essentially there are these two kinds of graphs that have appeared in multiple areas of mathematics. One is the plabic graph and one is the web and before our work, they were kind of viewed as separate objects. Starting with our visit to ICERM in May 2023 and continuing with this visit, we’ve been working on a series of papers in which we realize these objects are in fact very closely related to each other.

Striker: This week specifically, some of us were working on finishing [paper number three](#) in the series. Basically, there’s a two-parameter family of things where we’re trying to draw graphs for tableaux...what we did last May extended prior work people had done on three-row tableaux to the four-row case, and this relates over on the algebraic side to representation theory and special linear group and things like that. So some of us were editing paper number three and we finished, so that was exciting. And then a couple of us were writing some really fantastic code to help us do computational exploration, trying to help us come up with some moves on these graphs with a really nice user interface.

Swanson: There’s one particularly powerful routine which I spent a good chunk of time implementing. It basically finds all solutions to a large series of constraints. For our project we want to satisfy all these various constraints, and in large families of cases we have shown that, yes, you can. With further progress they get complicated enough that the code really is needed to help see what these solutions look like and really what’s going on, and that’s been very helpful. For instance, a few days ago after the algorithm was implemented I ran it on several thousand examples and then the seven hundredth that it checked was the first one that we couldn’t understand in some sense. So that gave us a new rule for these moves, and we never would’ve been able to check all those seven hundred cases and we would’ve basically tried to prove that the thing we wanted to be true is true using the rules that we have, and we would’ve gotten stuck because it’s not true...There’s another more complicated exchange relation that’s unlike any of the ones that’ve appeared in the web literature before, but with this approach we did find it.

ICERM: So it really helps expedite the process?

Swanson: It makes it possible. Like, there was a gap of about twenty years from the two-row case to the three-row case. And then there was a gap of thirty or so years from the three-row to us and the four-row. And I think a good chunk of why we were even able to do the four was because of the computer helping us to explore. ■

PROBABILITY PUZZLE

Contributed by Peter Francis, Evita Nestoridi, and Peter Winkler

Check you answer to the problem posed in [ICERM’s Fall 2023 Newsletter!](#)

You and your friend run a “random race” as follows. A continuous probability distribution on the real line is chosen. You both begin at 0. At time i , you and your friend independently draw from the distribution, and each of you advances by that distance. After n such moves, you win a valuable prize provided you not only win the race but lead after every step. What distribution is best for you, and what then is your probability of getting the prize?

Solution: You’re right—no matter what distribution you chose! Yes, the answer is the same, independent of the distribution. One way to show this is to observe that every generic set of n real numbers has the same number of signed permutations all of whose partial sums are positive, because that number doesn’t change when you move one of the reals up or down.

To determine what actually is the probability of leading all the way, pick the exponential distribution for your random race. “Memorylessness” leads you to the probability that a $2n$ -step random walk on the number line never dips below 0, namely $\binom{2n}{n}/4^n$.

For details and an open problem, check out [“Leading All the Way”](#) by Peter Francis, Evita Nestoridi and Peter Winkler!



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